

# popquiz 11 telescope ⚡

⚠ This is a preview of the published version of the quiz

Started: Mar 22 at 7:28pm

## Quiz Instructions



Question 1 12.5 pts



As discussed in class. This huge radio telescope is in



China



Peru

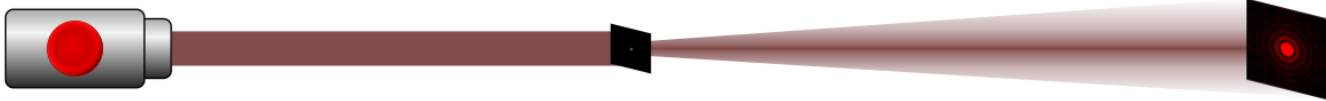
☐ US

☐ Argentina



### Question 2 12.5 pts

The property of light (or any wave) illustrated below



is called



reflection



diffusion



diffraction

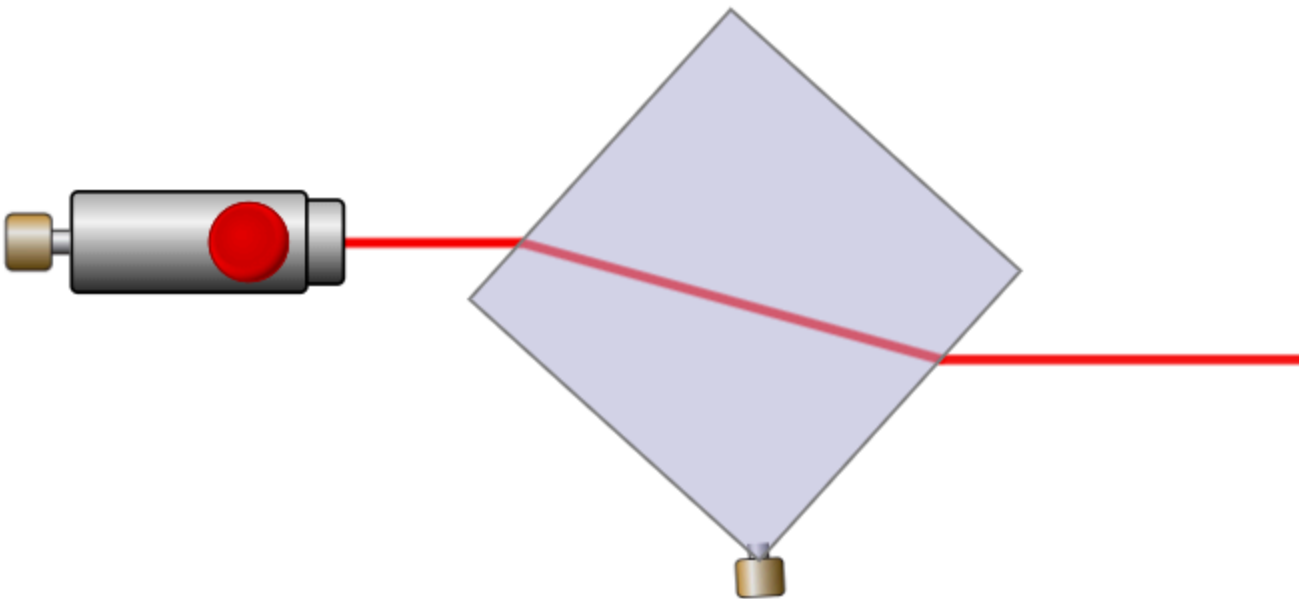


refraction



### Question 3 12.5 pts

The property of light illustrated below (there is a piece of glass)



is called



refraction



diffraction



reflection



diffusion



## Question 4 12.5 pts

The angular resolving power (or resolution) of a telescope. For the same distance. For the same object . Is better for



radiowave (collected from object)



blue light (collected from object)



red light (collected from object)



I didn't pay attention in class. and was wearing earphones too. too.



## Question 5 12.5 pts

The size of a human lens is about 6mm. The size of a typical telescope is 6m (ratio is 1000). The brightness of the same object observed with the telescope versus eye increases by a factor of : (slide 3. )

(the number of photons collected increase by a factor of ...)



I space out in class



1 billion (1,000,000,000)



1.000



1 million (1,000,000)



## Question 6 12.5 pts

The RQ-4 Global Hawk, a high-altitude surveillance drone, operates at **60,000 feet (18,288 meters)**. It uses the KFIR camera system with a **64mm (0.064m) lens diameter**. Assume visible light with a wavelength of **550 nm (0.00000055m or  $5.5 \times 10^{-7} \text{m}$ )**



**Calculate the minimum resolvable ground feature size using the formula from slide 16 (lambda is the wavelength)**

$$\text{Resolution} = \left( \frac{\lambda}{\text{Lens Size}} \right) \times \text{Altitude}$$

**hint: use only meters. Divide wavelength by size first. Then multiply by altitude**

☐

about 10 miles

☐

about 15 cm (about 6 in)

☐

about 7m (about 8 yards)

☐

about 5 cm (about 2 in)

☐

Question 7 12.5 pts

*Building on the previous question.*

*Which optical filter would maximize resolution for imaging ground features?*

*That is which color will give the best resolution?*

☐

Whaaaaaaaaat?

☐

yellow

☐

blue

☐

red



## Question 8 12.5 pts

This famous group of stars (asterism is



bootes



the summer triangle



the big dipper



Cassiopeia



## Question 9 5 pts

Which of those waves will diffract more (spread around corners).



visible light



x-rays



radio waves



UV light



Question 10 5 pts

Which color will refract more (bends more when going through glass)



radio wave



blue



red



green

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Not saved

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